

Taichi Kamei

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Embedded systems & Hardware engineer who takes projects from 0 to 1 under real physical constraints

EDUCATION

4th Year Engineering Physics

Expected Graduation - Dec 2028

University of British Columbia | Vancouver

Relevant Courses: Electric Circuit Analysis, Signals and Systems, Software Construction, Digital Systems and Microcomputers, Thermodynamics, Statistical Mechanics, Quantum Mechanics, Applied Linear Algebra

Dean's Honor List 2026

EXPERIENCE

Electronics & Controls Co-op

Sept 2026 - Present

LUNR Aerospace | Toronto

- Developing state machine in NI LabVIEW for 60kN rocket engine combustion control system, integrating multiple systems with temperature, pressure, load cell sensors and valves for different fluid
- Working on a modular Thrust Vector Control (TVC) firmware framework in C++, initially using for Simulink simulation and later ported on a flight controller hardware
- Using Quaternion and Extended Kalman Filter on IMU data for the TVC PID control

Firmware & Hardware Engineer Co-op

May - Aug 2026

UBC Blusson Quantum Matter Institute | Vancouver

- Initially started as a PCB validation role for the 4-channel voltage controlled biasing current source PCB used for the superconducting nanowire single photon detector (SNSPD)
- Recognized the need for firmware development, designed an active object architecture in C/C++ on ESP32 with ESP-IDF, FreeRTOS, dual-core task isolation, inter-core queue communication, and event coordination, achieving deterministic concurrent per-channel control
- Ported Analog Devices no-OS 24-bit Sigma-Delta ADC and 16-bit DAC drivers to ESP-IDF, implemented SPI platform abstraction layer with manual CS toggle for 5 SPI slaves and mixed SPI modes on a 10MHz shared bus
- Diagnosed ADC garbage reading due to SPI CPOL mismatch on different SPI modes (DAC: Mode 1, ADC: Mode 3), resolved the issue by adding an NOP transaction before actual transaction
- Developed hysteresis V-I sweep mode with 10mV step size and steady-state biasing mode with 10nA precision
- Designed calibration mode using Non-Volatile Memory, enabling runtime calibration lasting across power-cycle
- Debugged 3.3V LDO undervoltage by oscilloscope scoping and datasheet comparison, applied hardware workaround and PCB layout revision on Altium
- Tested time domain DAC and ADC drift characteristics by logging both PCB and commercial SMU measurement for 24 hours using Raspberry Pi
- Designed 8-channels co-planar waveguide PCB on Altium for 5GHz signals at 0.8 Kelvin cryogenic condition

PROJECTS

Continual Learning for Robotic Manipulation

Sept 2026 - Present

- Capstone project. Investigating the existing Continual Learning method for robotic arms to learn tasks sequentially and retain previously learned skill
- Will be replicating multiple representative SOTA robot-learning methods and evaluate on simulation platform
- Hoping to deploy and evaluate of selected SOTA and CL methods on a real robot at the end of the Capstone project

Drone Flight Controller & Power Board

Feb 2026 - Present

- Designed a 4-layer flight controller PCB on KiCAD, integrating ESP32-S3-Mini-1U, BMS, IMU, barometer, and a RF transceiver using SX1261 chip
- Implemented power regulation stage for 4S LiPo input with 5V/5A buck converter, 3.3V LDO for sensors and MCU, and a split ground for minimizing noise from high current 15V ESC line
- Used Dshot one-wire communication protocol for signaling commands to 4 independent ESC
- Designed a 4-layer GPS-module & magnetometer PCB fitting within 260mm by 260mm
- Redesigning flight controller PCB and power board PCB as separate boards for space optimization

Autonomous Clue Detecting Robot

Nov 2025

- Developed an autonomous vehicle in Gazebo and ROS 1 using OpenCV for PID control driving, obstruction avoidance, and CNN clue detection
- Designed and implemented a finite state machine (FSM) capable of transitioning its states for moving obstruction detection, clue board detection, and drive under various track surface conditions
- Optimized a PID algorithm for precise center-lane driving using dual side lines instead of a typical single line-following, overcoming camera center offset from the geometric center of the lane
- Created a Qt5 controller GUI which integrates simulation controls and launch of two ROS node scripts, boosting team productivity by centralizing all controls into a single window which normally takes 4+ separate terminals

Autonomous Payload Retrieving Robot

June - Aug 2025

- Designed a 3-DOF four-bar-linkage arm and arm base on Onshape and fabricated components using 3D printer, water jet cutter, and laser cutter
- Sized appropriate servo motors for the arm from maximum payload and bending moment calculations
- Implemented magnetic encoder-based rotational control for the arm base, achieving accuracy of less than 1° error
- Designed a custom 2-layer I2C multiplexer and I2C buffer PCB on KiCAD to solve peripheral address conflicts and signal degradation
- Prototyped a payload detection algorithm in Python on Raspberry Pi using two 2D LiDAR, generating depth and reflectance map in 7Hz. Implemented it in C++ on ESP32 with 15 Hz real-time detection

SKILLS

Software	C/C++, Python, LabVIEW, Rust, Java, Assembly, Simulink, VHDL, Linux, Git, CMake
Embedded	ESP-IDF, FreeRTOS, Arduino, Raspberry Pi, ROS, Gazebo, FPGA, I2C, SPI, UART, DAQ
Electrical	Kicad, Altium, LTSpice, Soldering, Oscilloscope, Electrometer, DMM, SMU, Logic Analyzer
Mechanical	Onshape, Siemens NX, 3D printing, Laser Cutting, Water Jet Cutting, Drill Press, Caliper